NPTEL ONLINE CERTIFICATION COURSES



Course Name: Software Project Management

Faculty Name: Dr. Durga Prasad Mohapatra

Department : Computer Sc. & Engg. NIT Rourkela

TOPIC: Project Monitoring and control



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CONCEPTS COVERED

- Gantt charts
- Slip chart
- Time line chart
- Cost monitoring



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Visualizing Progress

- Having collected data about project progress, a manager needs some way of presenting that data to greatest effect.
- Some methods of presenting a picture of the project and its future.
  - Methods that provide a static picture, a single snapshot (such as Gantt charts)
  - Methods that try to show how the project has progressed and changed through time (such as timeline charts)





Gantt chart

- Gantt chart has been named after its developer Henry Gantt.
- A Gantt chart is a form of bar chart.
- The vertical axis lists all the tasks to be performed.
- The bars are drawn along the y-axis, one for each task.
- Gantt charts used in software project management are actually an enhanced version of the standard Gantt charts. In the Gantt, each bar consists of a unshaded part and a shaded part.





Gantt chart

- The shaded part of the bar shows the length of time each task is estimated to take.
- The unshaded part shows the slack time or lax time.
- The lax time represents the leeway or flexibility available in meeting the latest time by which a task must be finished.





Gantt chart

- A Gantt chart is a special type of bar chart where each bar represents an activity. The bars are drawn along a time line. The length of each bar is proportional to the duration of time planned for the corresponding activity.
- Gantt chart representation of a project schedule is helpful in planning the utilisation of resources, while PERT chart is useful for monitoring the timely progress of activities.



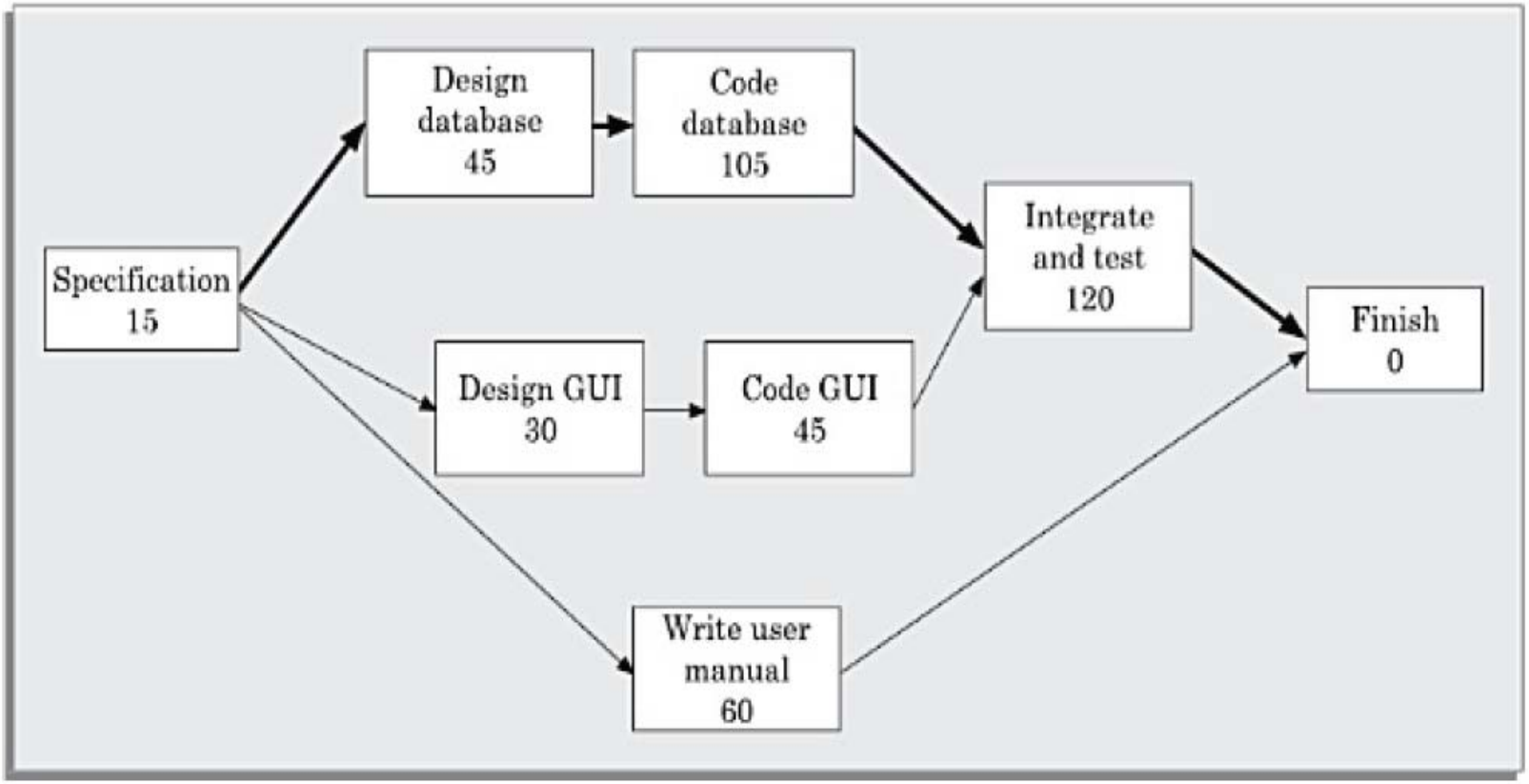


Gantt chart

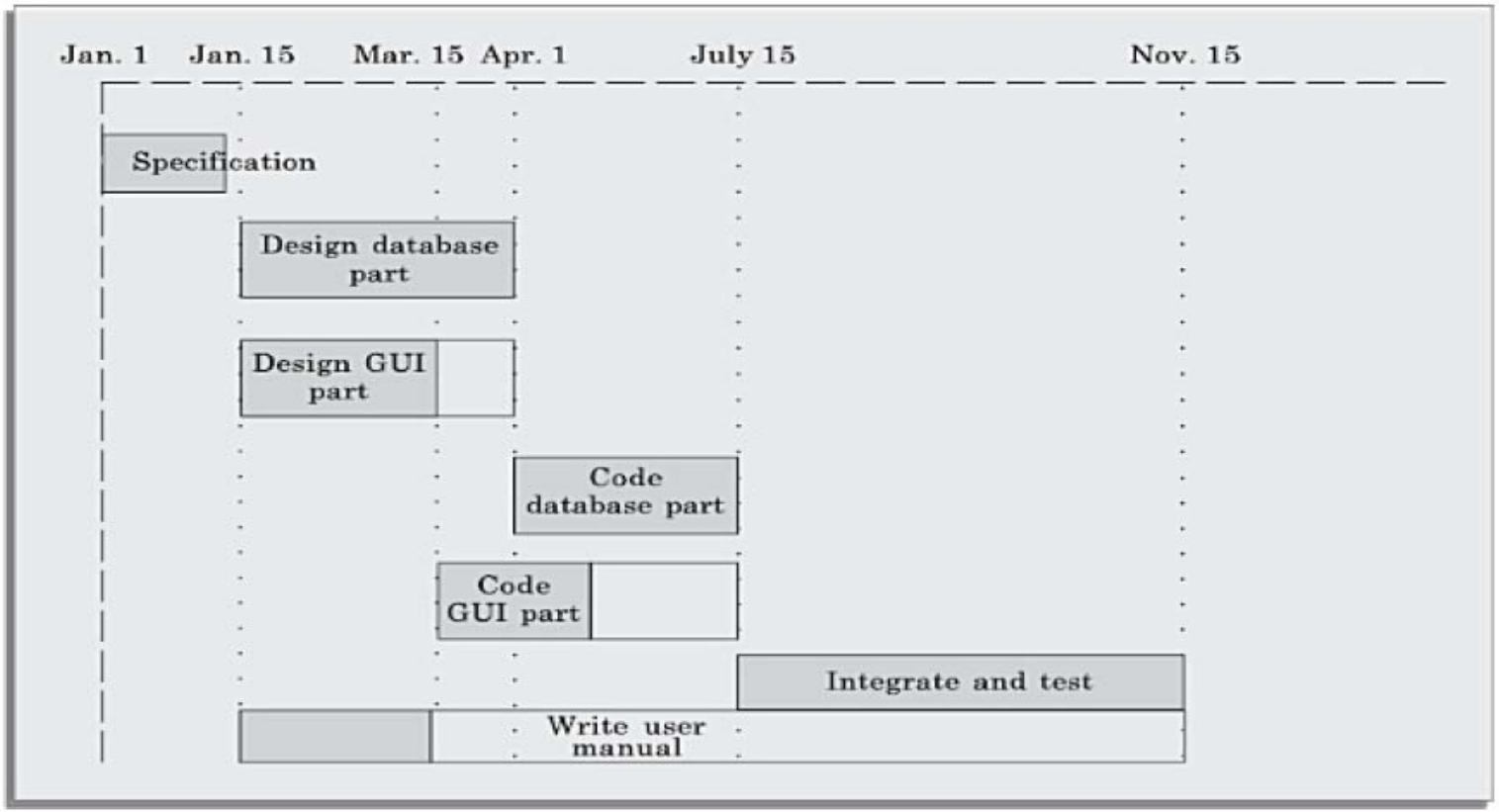
- Gantt charts are useful for resource planning (i.e. allocate resources to activities). The different types of resources that need to be allocated to activities include staff, hardware, and software.







Activity network representation of MIS problem



Gantt chart representation of the MIS problem

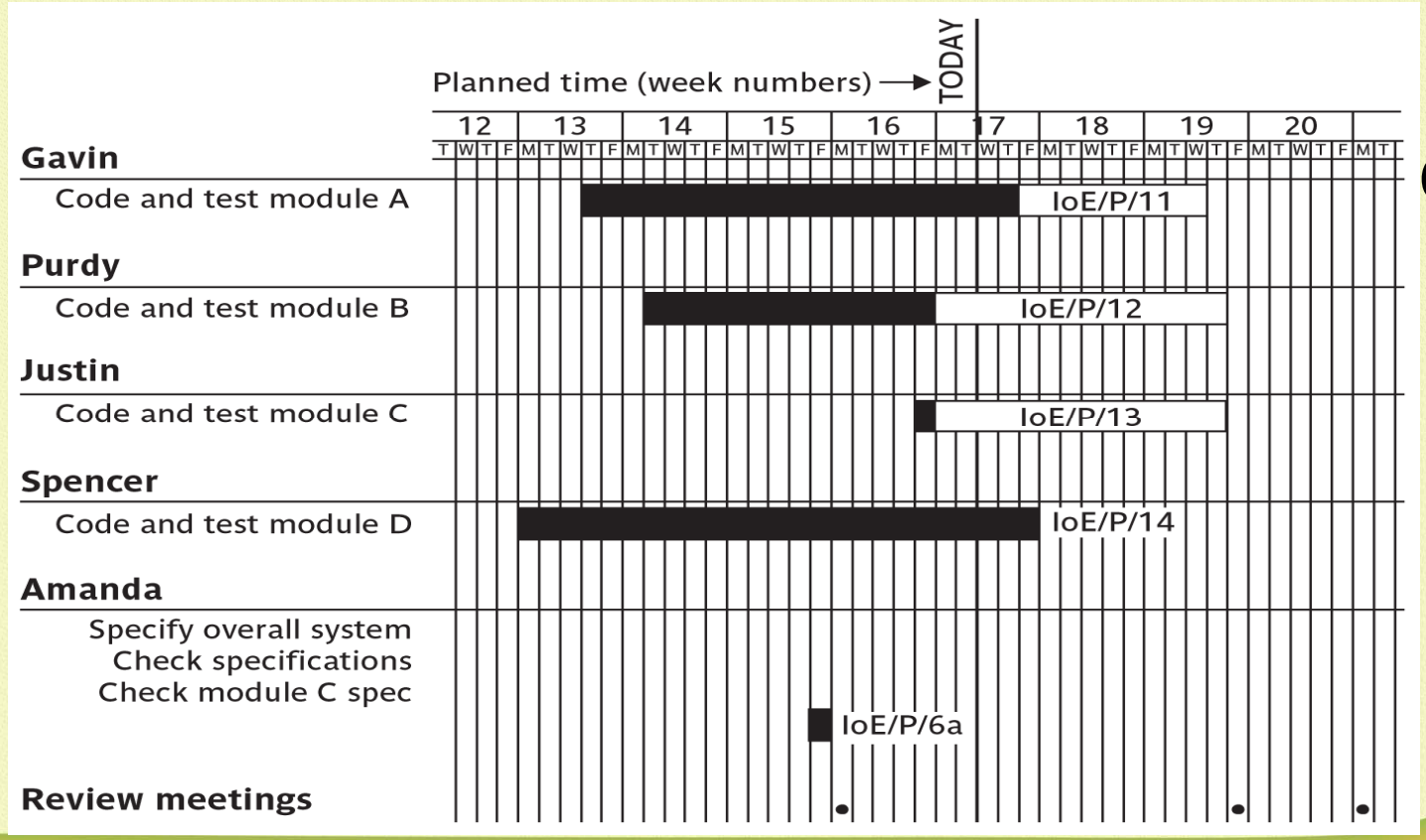
Gantt chart

- One of the simplest and oldest techniques for tracking project progress is the Gantt chart.
- This is essentially an activity bar chart indicating scheduled activity dates and durations, frequently augmented with activity floats (Activity float is the time by which an activity may be delayed without affecting any subsequent activity).
- Reported progress is recorded on the chart (normally by shading activity bars) and a ‘today cursor’ provides an immediate visual indication of which activities are ahead or behind schedule.





A Sample Gantt chart



Gantt chart cont …

- The figure shows a sample Gantt chart as at the end of Tuesday of week 17.
- ‘Code and test module D’ has been completed ahead of schedule and ‘Code and test module A’ appears also to be ahead of schedule.
- The coding and testing of the other two modules are behind schedule.





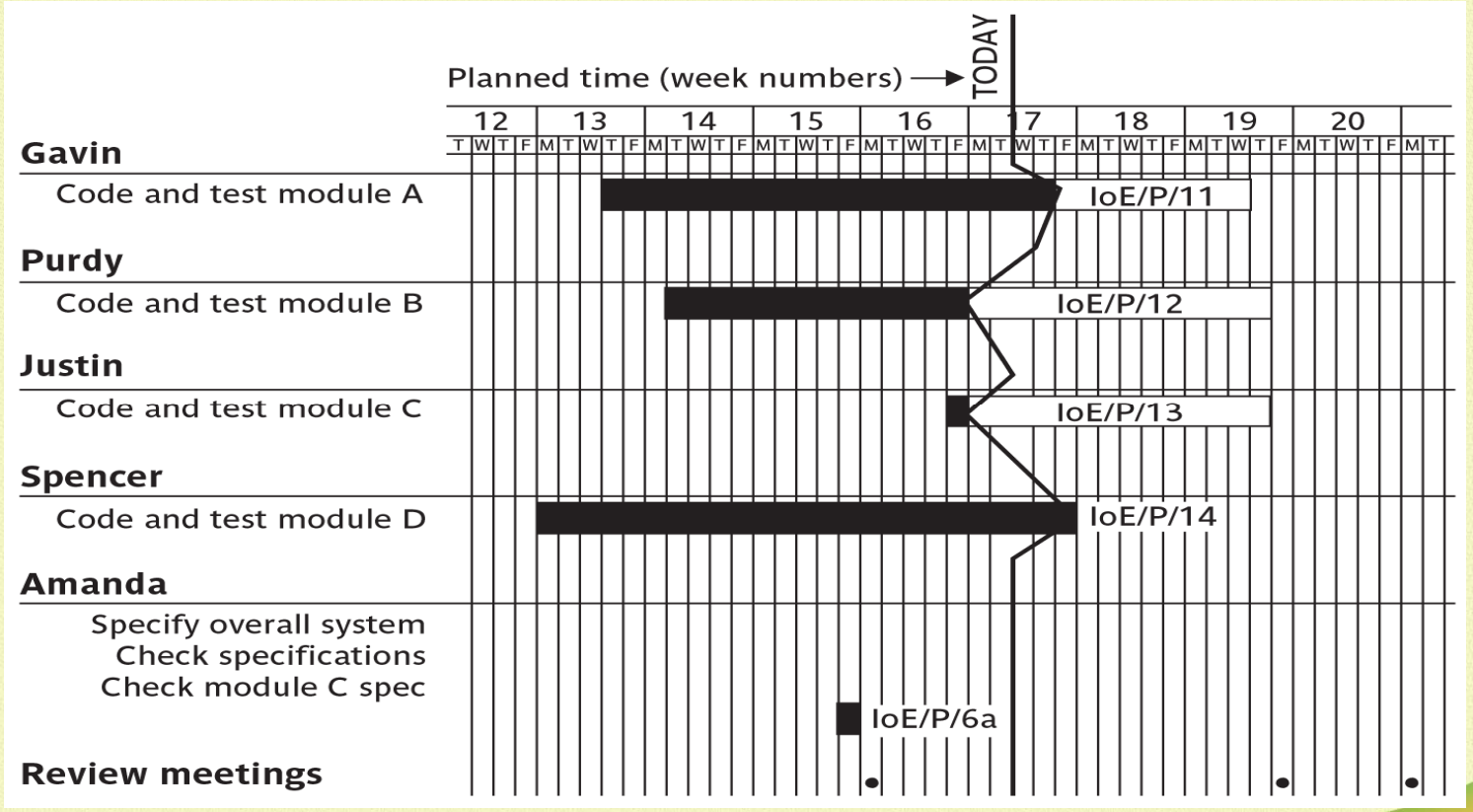
The slip chart

- A slip chart is a very similar alternative favoured by some project managers who believe
  - it provides a more striking visual indication of those activities that are not progressing to schedule,
    - the more the slip line bend, the greater the variation from the plan.
- Additional slip lines are added at intervals and, as they build up, the project manager will gain an idea as to whether the project is improving (subsequent slip lines bend less) or not.
- A very jagged slip line indicates a need for rescheduling.





A sample slip chart

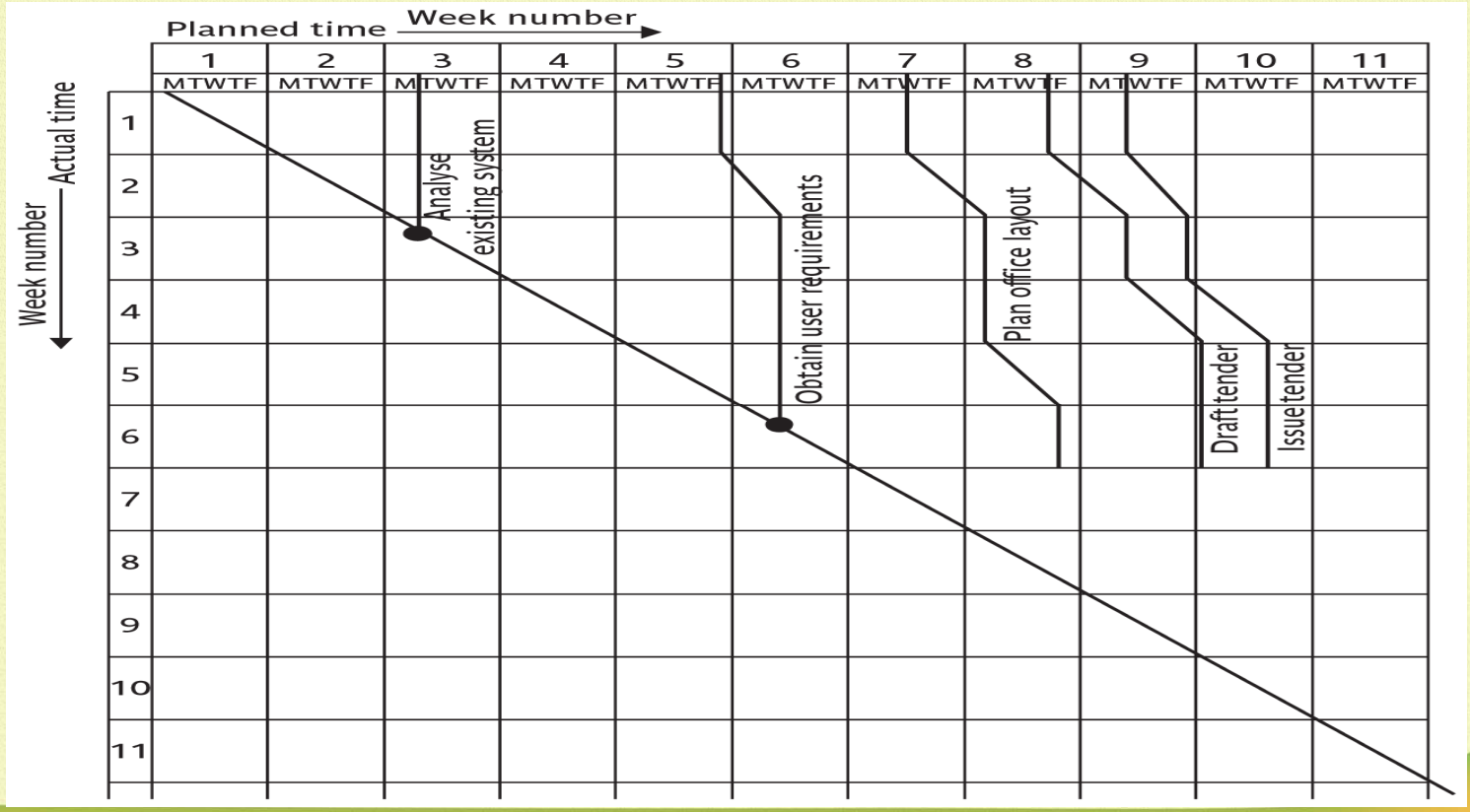


The timeline chart

- One disadvantage of Gantt chart and Slip chart is that they do not show clearly the slippage of the project completion date through the life of the project.
- Analysing and understanding trends in the project so far, allows us to predict the future progress of the project.
- For example, if a project is behind schedule because so far productivity has not been as high as assumed at the planning stage, it is likely that the scheduled completion date will be pushed back even further unless action is taken to compensate for or improve productivity.
- The timeline chart is a method of recording and displaying the way in which targets have changed throughout the duration of the project.







A sample timeline chart





The timeline chart cont …

- The figure shows a sample time line chart at the end of the sixth week.
- Planned time is plotted along the horizontal axis and elapsed time down the vertical axis.
- The lines meandering down the chart represent scheduled activity completion dates
  - At the start of the project ‘analyse existing system’ is scheduled to be completed by the Tuesday of week 3,
  - ‘obtain user requirements’ by Thursday of week 5
  - ‘issue tender’, the final activity, by Tuesday of week 9, and so on.





The timeline chart cont …

- At the end of the first week, the project manager reviews these target dates and leaves them as they are
  - lines are therefore drawn vertically downwards from the target dates to the end of week 1 on the actual time axis.
- At the end of week 2, he decides that 'obtain user requirements' will not be completed until Tuesday of week 6
  - he therefore extends that activity line diagonally to reflect this.
- The other activity completion targets are also delayed correspondingly.





The timeline chart cont …

- By the Tuesday of week 3, ‘analyse existing system’ is completed and the project manager puts a blob on the diagonal timeline to indicate that this has happened.
- At the end of week 3 he decides to keep to the existing targets.
- At the end of week 4 he adds another three days to ‘draft tender’ and ‘issue tender’.
- Note that by the end of week 6, two activities have been completed and three are still unfinished.
- Up to this point she has revised target dates on three occasions and the project as a whole is running seven days late.





The timeline chart cont …

- The timeline chart is useful both during the execution of a project and as part of the post-implementation review.
- Analysis of the timeline chart, and the reasons for the changes, can indicate failures in the estimation process or other errors that might, with that knowledge, be avoided in future.





Cost monitoring

- A project could be late because the staff originally committed have not been deployed
- In this case the project will be *behind time* but *under budget*
- A project could be on time but only because additional resources have been added and so be *over budget*
- Need to monitor both achievements and costs



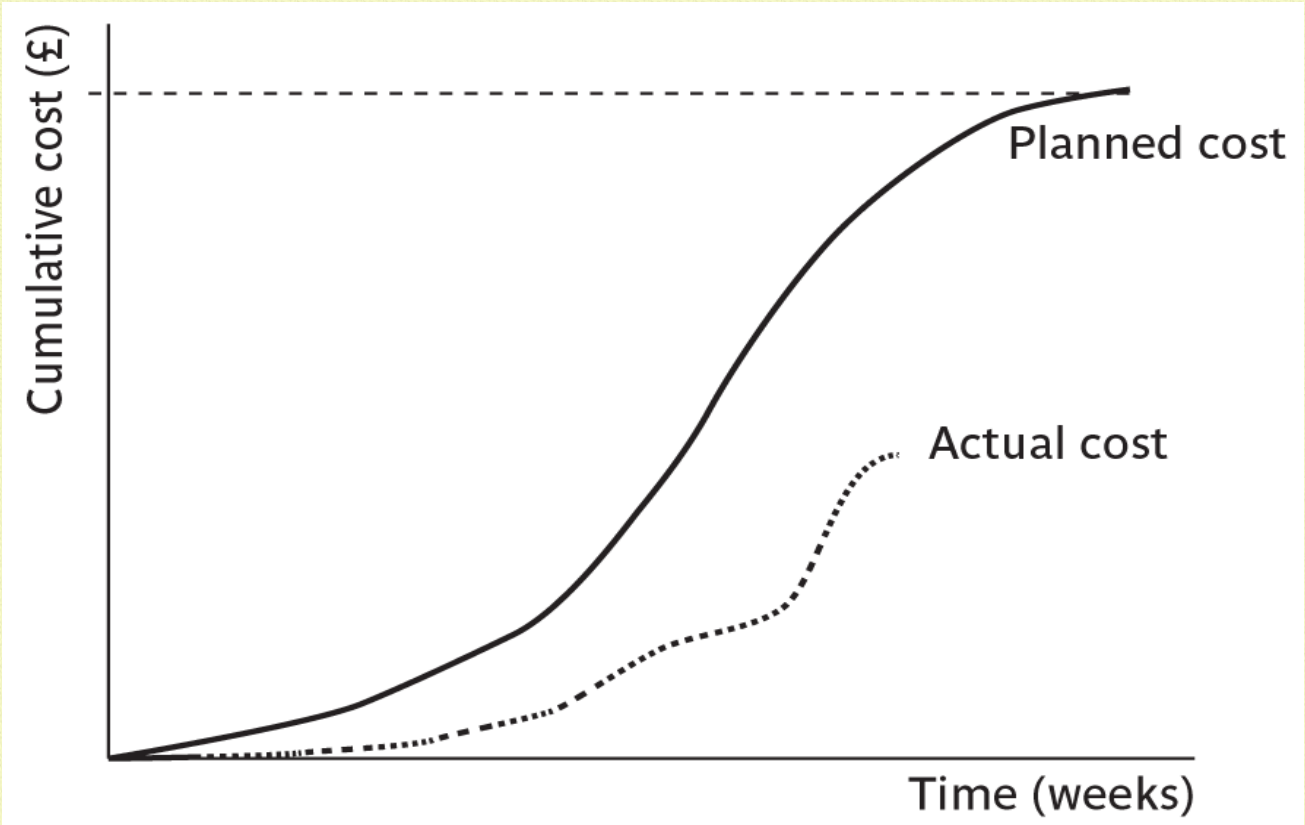


Cost monitoring cont ...

- Expenditure monitoring is an important component of project control, not only in itself, but also because it provides an indication of the effort that has gone into (or at least been charged to) a project.
- A project might be on time but only because more money has been spent on activities than originally budgeted.
- A cumulative expenditure chart provides a simple method of comparing actual and planned expenditure.
- By itself it is not particularly meaningful.
  - A project may run late or on time but the chart shows substantial costs savings.







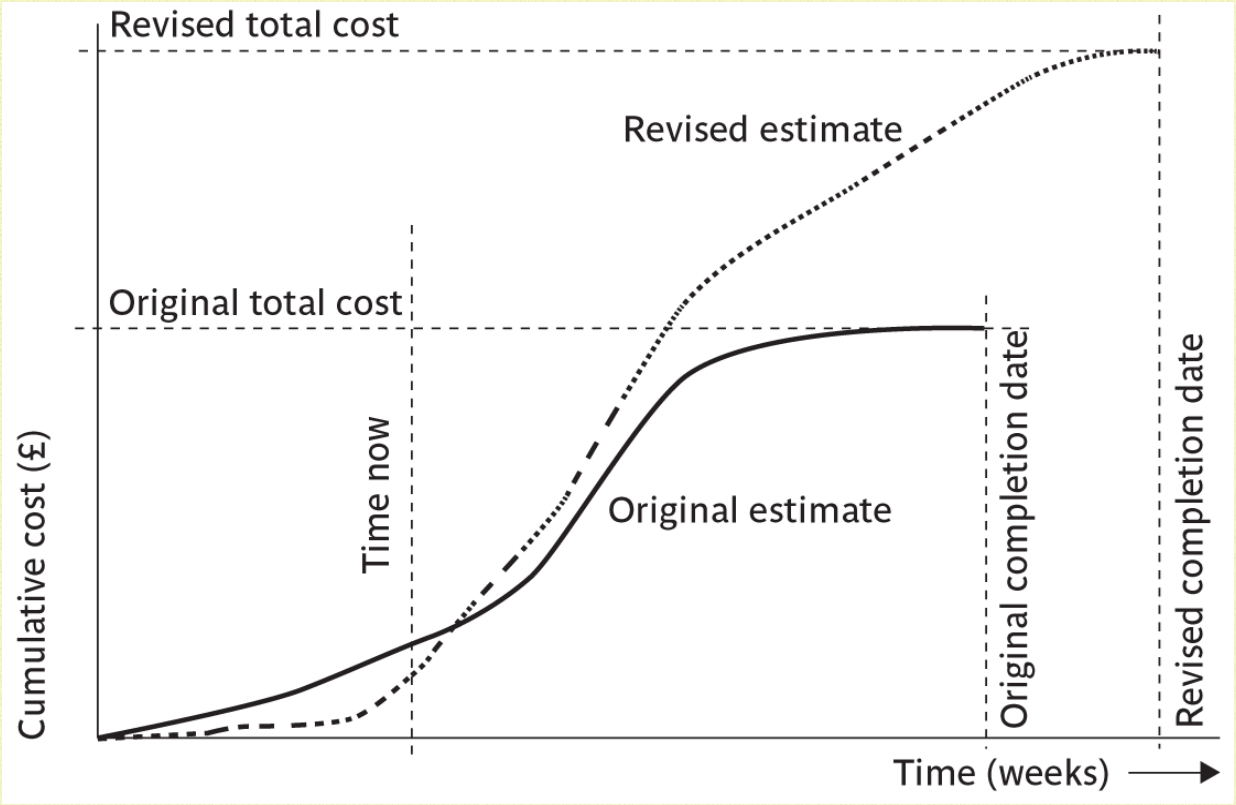
Tracking cumulative expenditure

Cost monitoring cont ...

- We need to take account of the current status of the project activities before attempting to interpret the meaning of recorded expenditure.
- Cost charts become much more useful if we add projected future costs calculated by adding the estimated costs of uncompleted work to the costs already incurred.
- Where a computer-based planning tool is used, revision of cost schedules is generally provided automatically, once actual expenditure has been recorded.







The cumulative expenditure chart can also show revised estimates of cost and completion date

Cost monitoring cont ...

- The figure illustrates the additional information available once the revised cost schedule is included
  - In this case it is apparent that the project is behind schedule and over budget.

Summary

- Discussed various ways to visualize the progress of a project such as
  - ✓ Gantt chart,
  - ✓ Slip chart and
  - ✓ Time line chart.
- Presented basic concepts of cost monitoring





References :

1. B. Hughes, M. Cotterell, R. Mall, *Software Project Management*, Sixth Edition, McGraw Hill Education (India) Pvt. Ltd., 2018.
2. R. Mall, *Fundamentals of Software Engineering*, Fifth Edition, PHI Learning Pvt. Ltd., 2018.






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